

PS-12 Cash Withdrawal Prediction

MVP – Complete Simple Summary

1. Project ka Main Purpose

Ye MVP fraud complaint ke baad money trail ko analyze karke suspicious ATM/Micro-ATM locations identify karega. System dekhega ki paisa kis route se gaya, kitni jaldi withdrawal hua aur location par pehle suspicious activity kitni thi.

Basic Flow

Victim → Account A → Account B → Account C → ATM/Micro-ATM → Risk Analysis → Risk Score → Top-3 Risky Locations

2. Input Data

- **Complaint data:** complaint ID, victim account, amount, timestamp.
- **Transaction data:** kis account se kis account mein paisa gaya aur kitne time mein gaya.
- **ATM/Micro-ATM data:** location, latitude/longitude aur normal daily activity/baseline.
- **Fraud chains:** 20–30 fake fraud chains ke saath kuch normal/legitimate records bhi.

MVP mein real sensitive police/bank data nahi; synthetic/fake data use hoga.

3. System kya Analyze Karega?

Signal	Simple Meaning
Conversion Speed	Complaint se withdrawal tak kitna time laga. Kam time = higher risk.
Network Depth	Paisa kitne accounts se hokar gaya. Zyada hops = higher risk.
Area Risk	ATM/area mein pehle kitni fraud-linked withdrawals hui.
Baseline	ATM ka normal daily transaction volume, taaki busy ATM ko galat tarah se risky na maana jaye.

4. Risk Score

In signals ko combine karke 0–100 ka Risk Score banega. PDF ka illustrative formula hai:

Score = (Network Risk × weight) + (Area Risk × weight) + (Time Risk × weight) – (Baseline × weight)

Example starting weights 0.4, 0.3, 0.2, 0.1 ho sakte hain. Ye validated weights nahi hain; presentation mein inhe illustrative/starting values ke roop mein batana hai.

Score	Category
0–40	Low
41–65	Medium
66–85	High
86+	Critical

Final output: sabhi locations ko high-to-low sort karke Top-3 risky locations dikhayi jayengi.

5. Map / Dashboard

- Website par map hoga.
- ATM locations ko H3 hexagons mein convert kiya jayega.

- Risk ke according areas highlight honge: Critical, High, Medium.
- Location par click/hover karne par Risk Score aur reason dikhega.
- Example: Risk 89 – Reason: high network fan-out + fast conversion.
- Relevant time-window bhi display kiya ja sakta hai.

Suggested tools: Python, h3-py, Streamlit; optional Folium + Flask.

6. End-to-End Demo

New Complaint Simulate Karo button → complaint process → features calculate → risk score → map update → Top-3 risky locations highlight.

Demo se pehle 2–3 different fake complaints par test aur ek team dry-run karna hai, taaki presentation ke waqt system crash na ho.

7. Future Roadmap

MVP	Future / Production
Synthetic data	Real complaint feeds
Simple weighted formula	Advanced ST-GNN model
Streamlit dashboard	Real-time streaming
Basic map/ranking	Kafka/Flink + Neo4j + real-time alerts

README mein installation, run instructions aur team contribution likhna hai.

8. Security

- Real Aadhaar/phone number jaise sensitive data use nahi karna.
- Fake/hashed identifiers use karna.
- RBAC aur basic security points presentation mein mention karna.
- Accuracy percentage tabhi claim karna jab synthetic data par actually calculate ki gayi ho.

9. 24-Hour Hackathon Plan

Time	Task
0–2 hr	Roles divide + fake data
2–6 hr	Features + risk formula
6–11 hr	Map/Dashboard
11–15 hr	Integration + end-to-end testing
15–19 hr	UI polish + explanation + bug fixing
19–21 hr	Deck + README
21–23 hr	Demo video
23–24 hr	Final links/files check + submission

10. One-Line Project Pitch

“Our system analyzes suspicious money movement after a complaint and identifies high-risk ATM/Micro-ATM locations using transaction-network, timing and area-risk signals.”

Final Simple Understanding

Fraud complaint aayi → system money chain dekhega → paisa kitni jaldi move hua dekhega → kitne accounts se gaya dekhega → area/ATM ka previous risk dekhega → sab signals ko score mein convert karega → Top-3 risky locations map par dikhayega. Ye hackathon MVP decision-support prototype hai, proven real-world fraud detector nahi.